ST. ANDREW'S JUNIOR COLLEGE

Higher 1

PRELIMINARY EXAMINATION 2012

Name ______________________________ **Class** ________________

CHEMISTRY **8872/ 02**

PAPER 2 **10 September 2012**

Candidates are to answer Section **A** on the Question Paper. **2 hours**

Additional Materials: Writing paper
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **two** questions on separate writing paper.

At the end of the exam, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's use	
Section A	/ 40
B6	
B7	
B8	
Total	/ 80

This paper consists of **16** printed pages and **1** blank page.

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Section A

Answer **all** questions in the spaces provided.

End of Section A

1. Magnesium exists as three principal isotopes, ^{24}Mg, ^{25}Mg and ^{26}Mg.

(a) Use the following data to calculate the relative atomic mass of magnesium to **two** decimal places.

Isotope	Relative isotopic mass	Natural abundance (%)
^{24}Mg	23.99	78.99
^{25}Mg	24.99	10.00
^{26}Mg	25.98	11.01

$$\text{RAM} = \frac{78.99 \text{ x } 23.99 + 10\text{x } 24.99 + 11.01 \text{ x } 25.98}{100}$$

$= 24.31$

(b) Magnesium and phosphorus are both in period 3. Describe and explain the difference in size of their atomic radii. .

Proton number of P > Mg. Hence, nuclear charge of P>Mg. Both have the same number of quantum shells hence shielding effect is almost the same for Mg and P. Therefore effective nuclear charge of P > Mg. Valence electrons are more strongly attracted to the nucleus of P than Mg hence atomic size of P is smaller than Mg.

(c) Magnesium and phosphorus form compounds with chlorine. The chlorides of the two elements have different pH in water as shown in the table.

Compound in water	pH
$MgCl_2$	6.5
PCl_3	2

With the aid of balanced equations, describe and explain the differences in pH values.

$MgCl_2$ partially hydrolyses in water due to high charge density of Mg^{2+} to give slightly acidic solution

$[Mg(H_2O)_6]^{2+}$ (aq) $\leftrightarrow$ $[Mg(H_2O)_5(OH)]^{+}$ (aq) + H^{+} (aq)

PCl_3 is covalent, hydrolyses in water to form strong acids

$PCl_3 (l) + 3H_2O (l) \rightarrow H_3PO_3 (aq) + 3HCl (aq)$

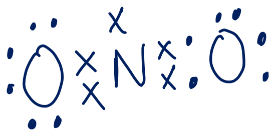
[Total: 6 m]

2. One of the gases responsible for acid rain, nitrogen dioxide, NO_2, could be produced in the car engine by the following reaction sequence.

$$\textbf{I} \quad N_2(g) + O_2(g) \rightarrow 2NO(g), \quad \Delta H = +180 \text{ kJmol}^{-1}$$

$$\textbf{II} \quad NO(g) + \tfrac{1}{2}O_2(g) \rightarrow NO_2 (g), \quad \Delta H = -57 \text{ kJmol}^{-1}$$

(a) Draw a dot-and-cross diagram of a molecule of NO_2. Hence, state its shape and bond angle.



Bent , <120°

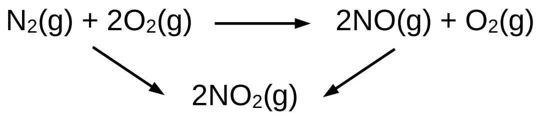
(b) Explain briefly why reaction **I** occurs in car engines despite the high activation energy, and how NO gas can be removed in the car exhaust by a catalytic converter.

The high temperature enables the reaction to proceed despite the high activation energy

It can be reduced to nitrogen gas by reacting with CO or unburnt hydrocarbons

or equations, eg: $CO + NO \rightarrow CO_2 + \tfrac{1}{2}N_2$

(c) Using the following energy cycle or otherwise, calculate the enthalpy change of formation of NO_2.



$\Delta H = +180 + 2(-57) = +66 \text{ kJmol}^{-1}$

$\Delta H_f = 0.5 \times 66 = +33 \text{ kJ mol}^{-1}$

[Total: 6 m]

3. The Haber process, is the nitrogen fixation reaction of nitrogen gas and hydrogen gas, over an enriched iron or ruthenium catalyst, which is used to industrially produce ammonia. The reaction is as follows:

$$N_2\,(g) + 3\,H_2\,(g) \rightleftharpoons 2\,NH_3\,(g)$$

(a) (i) Write an expression for the equilibrium constant, K_c for the reaction.

$$K_c = \frac{[NH_3]^2}{[N_2]\,[H_2]^3}$$

(ii) A mixture of N_2 and H_2 gases is heated in a 3.0 dm^3 closed container and the reaction allowed to reach equilibrium at 200 atm and 280 °C. The equilibrium mixture contains 0.80 mol of N_2, 0.50 mol of H_2 and 1.60 mol of NH_3. Calculate the value of K_c at 200 atm and 280°C, stating the appropriate units.

$$K_c = \frac{(\frac{1.6}{3})^2}{(\frac{0.8}{3})(\frac{0.5}{3})^3}$$

$= 230\ mol^{-2}\ dm^{6}$

(iii) State and explain how the position of equilibrium and the equilibrium constant would change if the pressure is increased to 400 atm.

When pressure increases, eqm will shift to the right with less moles of gaseous molecules to lower the pressure.
Since temp is the same, K_c remains the same.

(b) Experiment **A** was carried out starting with 2 mol of nitrogen and 6 mol of hydrogen at a 200 atm and 280°C. Curve **A** shows how the number of moles of ammonia present changed with time in experiment **A.**

Curves **B**, **C** and **D** refer to similar experiments, starting with the same amount of reactants but each with different conditions.

(i) Use Le Chatelier's principle to identify which one of the curves **B**, **C** or **D** represents an experiment carried out at the same temperature as experiment **A** but at a higher pressure. Explain your answer.

Curve B

At higher pressure eqm shift to the right where there are less moles of gas to decrease the pressure . Hence more ammonia is produced (or yield increases)

(ii) Suggest the change in conditions that would produce curve **C**?

A catalyst was added to the reaction.

(c) (i) Ammonia burns in oxygen to give nitrogen dioxide and steam. Write the equation which represents the enthalpy change of combustion of ammonia.

$NH_3 (g) + 7/4\ O_2 (g) \rightarrow NO_2 (g) + 3/2\ H_2O (g)$

(ii) Hence, use the following information to calculate a value for the enthalpy change of combustion of ammonia.

Enthalpy change of formation of nitrogen dioxide	-34 $kJmol^{-1}$
Enthalpy change of formation of steam	-241.83 $kJmol^{-1}$
Enthalpy change of formation of ammonia	-46 $kJmol^{-1}$

$\Delta H = (3/2 \times -241.83 + (-34)) - (-46)$

$= -351\ kJmol^{-1}$.

(iii) Using relevant information from the data booklet as well as your answer in c(ii), calculate a value for the bond energy of the bond between the nitrogen atom and the oxygen atom in nitrogen dioxide, assuming that the bond energy of both bonds are the same.

Bonds broken = 3 x 390 + 7/4 x 496 = 2038 $kJmol^{-1}$

Bonds formed = 3 x 460 + 2 x BE (N-O) = 1380 + 2 BE(N-O)

-350.75 = 2038 – 1380 – 2 BE(N-O)

BE(N-O) = +504.38

= +504 $kJmol^{-1}$

(iv) The combustion of 44 g of ammonia produces 89 g of nitrogen dioxide. Calculate the percentage efficiency of this reaction.

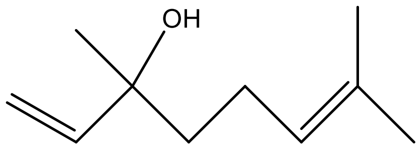
Amt of ammonia = 44/17 = 2.59 mol = amt of NO_2

Theoretical mass of NO_2 = 2.59 x 46 = 119.14 g

Percentage efficiency = 89/119.14 x 100 = 74.7%

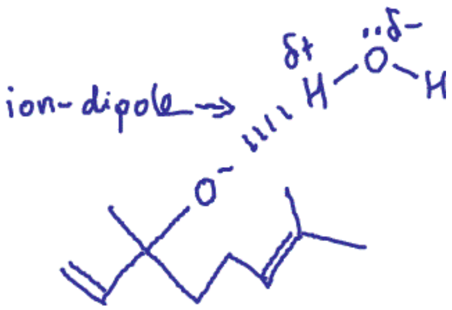
[Total: 13 m]

4(a) Lavender oil has long been used in aromatherapy. The scent has a calming effect which may aid in relaxation and the reduction of anxiety. Linalool is one of the primary components of lavender oil.



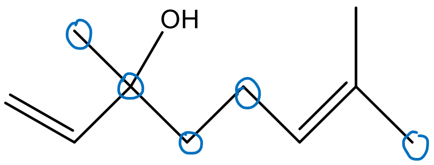
linalool

(i) Linalool reacts with Na metal to give compound **Y**. Explain with the aid of a diagram why compound **Y** is soluble in water.

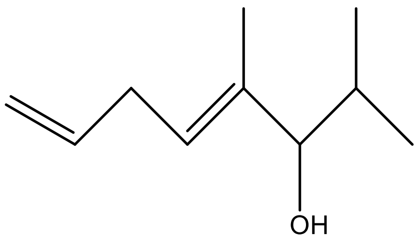


Y dissolves in water as it can form ion-dipole interactions with water.

(ii) Mark on the structure of linalool all the carbon that are sp^3 hybridised.



(iii) Suggest a simple chemical test to distinguish between linalool and its isomer, **Z**.

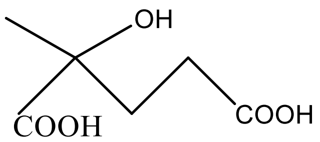


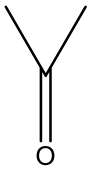
Z

Reagents and conditions: aq H_2SO_4 , aq $K_2Cr_2O_7$, heat

Observations: Z turns orange dichromate green. Orange solution remains for linalool

(v) Draw the structures of the organic products that are formed when hot acidified $KMnO_4$ is added to linalool.

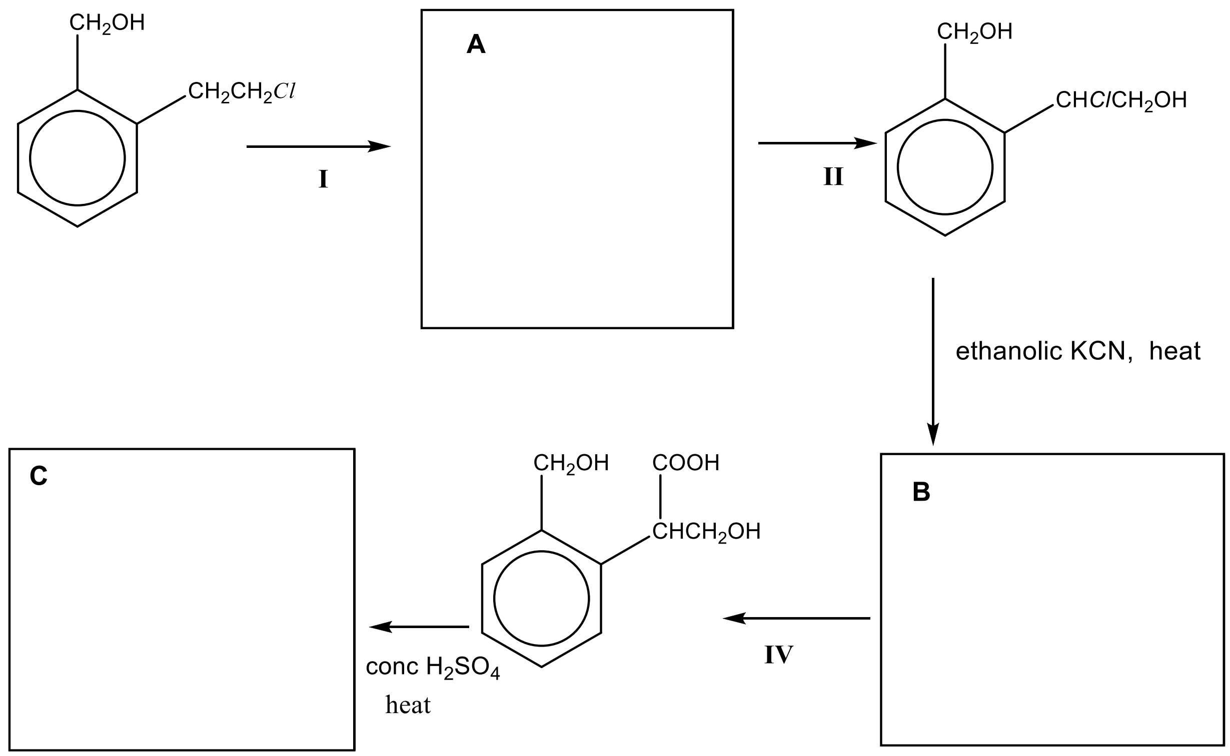




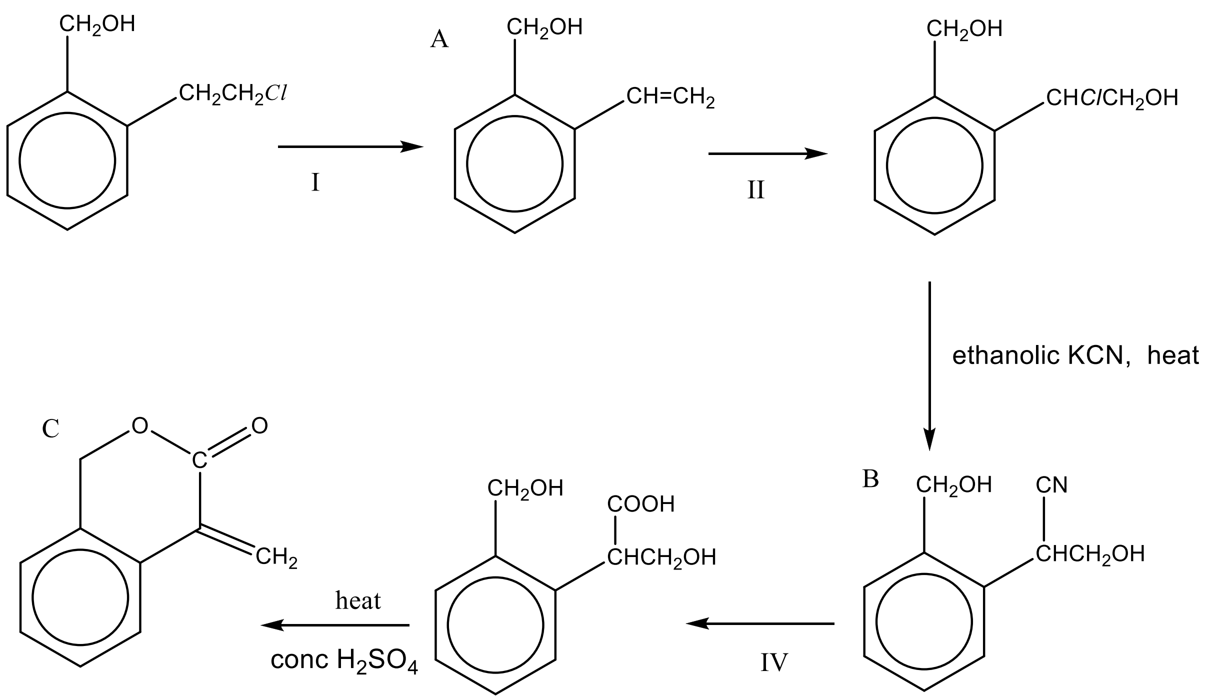
1m each

[Total: 7 m]

4(b) The reaction scheme below shows the synthesis of a sweet-smelling compound **C**.

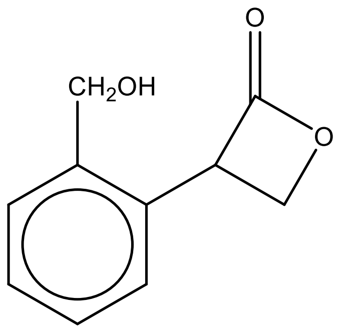


CH2OH
CH2CH2Cl
I
A
II
CH2OH
CHClCH2OH
ethanolic KCN, heat
C
CH2OH
COOH
CHCH2OH
B
conc H2SO4
heat
IV



CH2OH
CH2CH2Cl
I
A
CH2OH
CH=CH2
II
CH2OH
CHClCH2OH
ethanolic KCN, heat
C
O
C
O
CH2
heat
conc H2SO4
CH2OH
COOH
CHCH2OH
IV
B
CH2OH
CN
CHCH2OH

or



(i) Draw the structures of A to C in the spaces provided

(ii) Give the reagents and conditions for Steps **I**, **II** and **IV**.

I: ethanolic KOH, heat

II: Cl_2 (aq)

III: aq H_2SO_4, heat

(iii) Name the type of reaction for Steps **II** and **IV**

II : addition

IV: hydrolysis

[Total: 8 m]

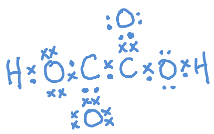
[Turn Over

Section B

Answer 2 out of 3 questions.

5. Ethanedioic acid, $(COOH)_2$, is often used as a cleaning agent for the removal of rust.

(a) Draw a dot-and-cross diagram to show the arrangement of electrons in the $(COOH)_2$ molecule. Hence, suggest the bond angle about either carbon atom.



Bond angle: 120°

[2]

(b) Ethanedioic acid is a *weak Bronsted acid*.

What do you understand by the term in italics? **[2]**

Bronsted acids are proton donors.

Weak acids are acids that dissociate partially in water to give H^+.

(c) A solution containing $(COOH)_2$ and its salt, $HOOC\text{-}COO^-Na^+$, acts as a buffer solution.

(i) What do you understand by the term *buffer solution*?

A buffer solution is a solution whose pH remains almost constant when small amounts of acid or base are added to it.

(ii) Write ionic equations to show how this solution reacts with a small amount of

I H^+ (aq) ions,

II OH^- (aq) ions.

$HOOC\text{-}COO^- + H^+ \rightarrow HOOC\text{-}COOH$

$HOOC\text{-}COOH + OH^- \rightarrow HOOC\text{-}COO^- + H_2O$

[3]

(d) Ethanedioic acid can be oxidised by acidified potassium manganate(VII).

When a 0.32 g impure sample of ethanedioic acid was reacted with an excess of acidified potassium manganate(VII), the carbon dioxide gas produced was bubbled into an excess of $Ba(OH)_2$ solution. The white precipitate formed was dried and was found to weigh 0.788 g.

5. (d) (i) Write an ion-electron equation for the formation of CO_2 from ethanedioic acid.

Hence write a balanced equation for the reaction between ethanedioic acid and acidified potassium manganate(VII).

$H_2C_2O_4 \rightarrow 2\ CO_2 + 2H^+ + 2e$

$5\ H_2C_2O_4 + 2\ MnO_4^- + 6\ H^+ \rightarrow 10\ CO_2 + 2\ Mn^{2+} + 8\ H_2O$

(ii) State the identity of the white precipitate formed from the reaction of CO_2 and $Ba(OH)_2$.

$BaCO_3$

(iii) Calculate the percentage purity of ethanedioic acid in the sample.

amount of $BaCO_3$ = 0.788/(137 + 12 + 48) = 0.004 mol

$BaCO_3 : CO_2 : H_2C_2O_4$ = 1 : 1 : 0.5

amount of $H_2C_2O_4$ = 0.5 × 0.004 = 0.002 mol

mass of $H_2C_2O_4$ = 0.002 × (2 + 12 x 2 + 64) = 0.18 g

% purity = 0.18/0.32 × 100% = 56.3 %

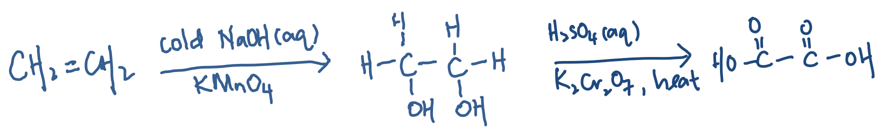
[7]

(e) Give the reagents, conditions and the structure of all intermediates to show how the synthesis of ethanedioic acid would be carried out from

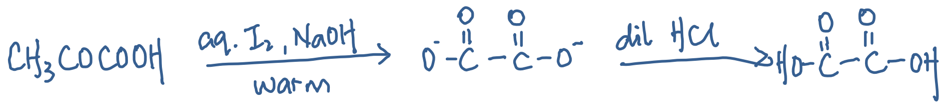
(i) $CH_2{=}CH_2$

(ii) $CH_3COCOOH$

(i)



(ii)



[6]

[Total: 20]

6. **(a)** The rate of reaction between 2-bromo-2-methylpropane, $(CH_3)_3CBr$, and aqueous sodium hydroxide when heated under reflux is found to be independent of $[OH^-]$ but directly proportional to $[(CH_3)_3CBr]$. Hence the rate equation is

$$\text{Rate} = k[(CH_3)_3CBr]$$

(i) State the type of reaction above.

substitution

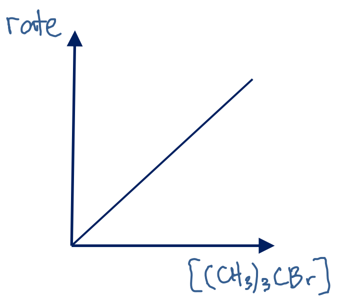
(ii) Write a balanced chemical equation for the reaction.

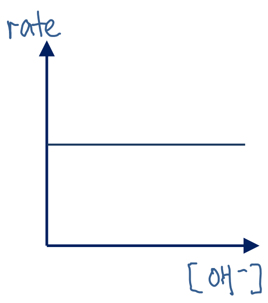
$(CH_3)_3CBr + NaOH \rightarrow (CH_3)_3COH + NaBr$

(iii) Sketch a graph of rate against concentration for

I $(CH_3)_3CBr$

II OH^-



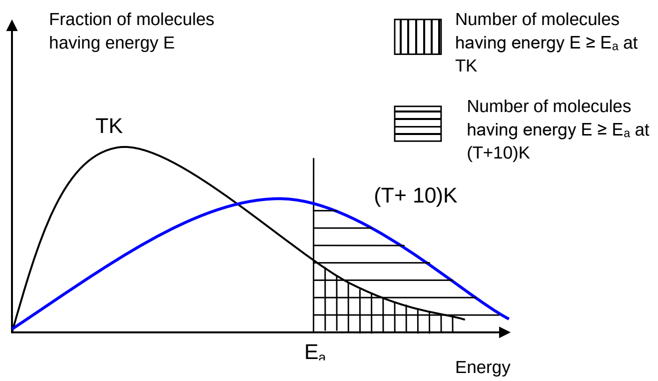


(iv) Predict and explain the relative rate of reaction if $(CH_3)_3CCl$ was used instead in the reaction above.

The rate will be slower.

C-C*l* bond is stronger as its bond length is smaller/ overlap of orbitals is more effective due to C*l* having a smaller atomic radius.

(v) With the aid of an appropriate sketch of the energy distribution curve, explain why an increase in temperature can speed up this chemical reaction.



When temperature increases,

- Kinetic energy of the molecules increases
- More molecules have energy greater than or equal to the activation energy
- Frequency of effective collision increases

[9]

(b) Straight chain alkanes undergo thermal cracking to give shorter chain hydrocarbons. When straight chain $C_{14}H_{30}$ undergoes cracking, compounds **P**, **Q** and **R** are formed in the ratio of 2 : 1 : 1.

(i) When 20 cm^3 of **P** is completely burnt in 150 cm^3 of oxygen gas, the resultant gaseous mixture occupies 110 cm^3 and has an average relative molecular mass of 38.5.

Deduce the molecular formula of **P**.

(All volumes are measured at room temperature and pressure.)

$C_xH_y + (x + y/4)\, O_2 \rightarrow x\, CO_2 + (y/2)\, H_2O$

The resultant gas comprises of unused O_2 and CO_2 produced.

Let the mole fraction of O_2 = a

32a + 44 (1 – a) = 38.55

12a = 5.5

a = 0.458

Volume of unused O_2 = 0.458 × 110 = 50.4 cm^3

Volume of CO_2 = 110 – 50.4 = 59.6 cm^3

x = 59.6/20 = 3

x + y/4 = (150 – 50.4)/20 = 5

y = 8

P : C_3H_8

Alternative working:

C_xH_y + (x + y/4) O_2 → x CO_2 + (y/2) H_2O

Vol of oxygen = 20 (x + y/4)

Vol of CO_2 = 20x

Vol of oxygen left = 150 - **20** (x + y/4)

150 - **20** (x + y/4) + 20x = 110

150 – 5y = 110

y = 8

$$\frac{20x}{110}(44) + \frac{150 - 20(x + y/4)}{110}(32) = 38.5$$

$$8x + \frac{4800 - 640x - 160y}{110} = 38.5$$

Subst y = 8

380x + 4800 – 640x -1280 = 4235

$x = 3$

P is C_3H_8

(ii) Using the following information, deduce the structure of **Q**.

- **Q** comprises 85.7% C and 14.3% H.
- **Q** exhibits geometric isomerism.
- 0.826 g of gaseous **Q** occupies 354 cm^3 at room temperature and pressure.

	C	**H**
Percentage	85.7	14.3
amount	85.7/12 = 7.142	14.3/1 = 14.3
ratio	1	2

Empirical formula = CH_2

amount of Q = 354/24000 = 0.01475 mol

Mr = 0.826/0.01475 = 56

$(CH_2)_n = 12n + 2n = 56$

$n = 4$

Molecular formula C_4H_8

Structure of **Q** : $CH_3CH{=}CHCH_3$

(iii) Hence deduce the molecular formula of **R**.

$C_{14}H_{30} \rightarrow 2\ C_3H_8 + C_4H_8 + \mathbf{R}$

R : $C_{(14-10)}H_{(30-24)} = C_4H_6$

(iv) 1 mole of **R** reacts with 2 moles of aqueous Br_2. Upon reaction with hot acidified $KMnO_4$, **R** produces only carbon dioxide and water.

Identify **R** and write an equation for the reaction of **R** and aqueous Br_2.

R : $CH_2=CHCH=CH_2$

$CH_2=CHCH=CH_2 + 2Br_2 + 2H_2O \rightarrow CH_2BrCH(OH)CH(OH)CH_2Br + 2HBr$

[11]

[Total: 20]

7. **(a)** Lemon juice contains citric acid which is a weak acid. The pH of lemon juice is 2.11.

$$C_6H_8O_7 \rightleftharpoons C_6H_7O_7^- + H^+$$

The concentration of citric acid in lemon juice is determined by titrating 25.0 cm^3 of the solution with 0.200 mol dm^{-3} of sodium hydroxide. The end-point was reached when 35.80 cm^3 of sodium hydroxide was added.

(i) Write an expression for the acid dissociation constant, K_a, of citric acid.

$K_a = [C_6H_7O_7^-][H^+] / [C_6H_8O_7]$

(ii) Calculate the concentration of citric acid in lemon juice.

amount of citric acid = amount of NaOH = 0.0358 × 0.200 = 0.00716 mol

[citric acid] = 0.00716/0.025 = 0.286 mol dm^{-3}

(iii) Hence prove that citric acid is a weak acid.

$[H^+] = 10^{-2.11} = 0.00776$ mol dm^{-3}

$[H^+] <<$ [citric acid]

citric acid dissociates partially, hence it is a weak acid.

(iv) Given that $[C_6H_7O_7^-] = [H^+]$, calculate K_a of citric acid, stating its units.

$K_a = [C_6H_7O_7^-][H^+] / [C_6H_8O_7] = (0.00776)^2 / 0.286 = 2.11 \times 10^{-4}$ mol dm^{-3}

(iv) Write an equation to show that the salt formed between citric acid and sodium hydroxide is alkaline.

$C_6H_7O_7^- + H_2O \rightleftharpoons C_6H_8O_7 + OH^-$

(v) State a suitable indicator for the titration.

phenolphthalein

[8]

(b) **(i)** Bauxite is an aluminium ore and is the main source of aluminium. To extract the aluminium in the ore, sodium hydroxide is used to dissolve aluminium oxide in the ore.

Give a balanced chemical equation for the reaction between sodium hydroxide and aluminium oxide.

$Al_2O_3 + 2\ NaOH + 3\ H_2O \rightarrow 2NaAl(OH)_4$

(ii) Another aluminium-containing compound is aluminium sulfate. A pure sample of aluminium sulfate was dissolved in water. When solid calcium carbonate was added into the solution, effervescence was observed.

Explain the reactions, giving equations where appropriate.

Al^{3+} has high charge density and hydrolyses in water to give an acidic solution.

$[Al(H_2O)_6]^{3+} + H_2O \leftrightarrow [Al(H_2O)_5(OH)]^{2+} + H_3O^+$

Calcium carbonate reacts with the acidic solution to give CO_2.

$CaCO_3 + 2\ H_3O^+ \rightarrow Ca^{2+} + CO_2 + 3\ H_2O$

or

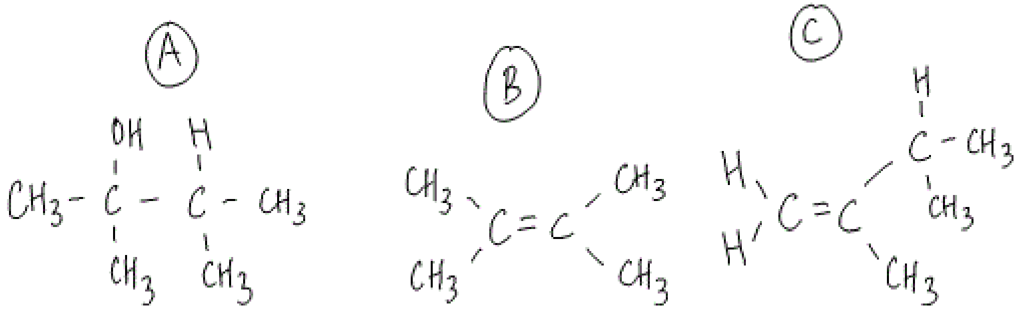
$CaCO_3 + 2\ H^+ \rightarrow Ca^{2+} + CO_2 + H_2O$

[4]

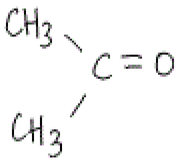
(c) Aluminium oxide can also be used in organic chemistry as a dehydrating agent. When aluminium oxide reacts with compound **A**, $C_6H_{14}O$, two isomers **B** and **C** are formed. When the two isomers are reacted with hot acidified potassium manganate(VII), isomer **B** gives compound **D**, C_3H_6O, as the only product. While isomer **C** gives compound **E** and effervescence is seen. Both compounds **D** and **E** give an orange precipitate with 2,4-dinitrophenylhydrazine and a yellow precipitate in alkaline aqueous iodine.

Deduce the structures of compounds **A – E**. Explain the chemistry of the reactions.

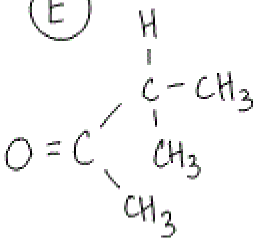
Observations	**Deductions**
When aluminium oxide reacts with compound **A**, $C_6H_{14}O$, two isomers **B** and **C** are formed.	Elimination **A** is an alcohol **B** and **C** are alkenes
When the two isomers are reacted with hot acidified potassium manganate(VII), isomer **B** gives compound **D**, C_3H_6O, as the only product.	Oxidation **B** gives only one product hence it is symmetrical about the C=C
While isomer **C** gives compound **E** and effervescence is seen.	**C** has C=C on a terminal carbon
Both compounds **D** and **E** give an orange precipitate with 2,4-dinitrophenylhydrazine	Condensation **D** and **E** are ketones
Both compounds **D** and **E** give a yellow precipitate in alkaline aqueous iodine.	Oxidation **D** and **E** have the structure CH_3CO-



D



E



[8]

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End of Paper 2